

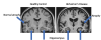
Klasifikasi Tingkat Keparahan Penyakit Alzheimer Berdasarkan Citra Otak

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<https://pub.com/indexsamudhuri/alzheimer-detection-gi>

Summary

Penyakit Alzheimer, suatu kondisi neurodegeneratif yang ditandai dengan hilangnya memori, fungsi kognitif, dan perilaku normal, merupakan salah satu penyebab utama demensia.



Penelitian ini menggunakan metode deep learning untuk mengklasifikasi tingkat keparahan penyakit Alzheimer berdasarkan citra MRI otak. Model yang dikembangkan mampu mengidentifikasi pola-pola yang berkaitan dengan penyakit Alzheimer, termasuk atrofi hipokampus, plak amyloid, dan perubahan struktural lainnya.

Deep Learning

1. Fine-Tuning



2. CNN Pipeline



3. Transfer Model Transformer



4. PyTorch Framework

- Easy-to-use deep learning framework
- Extensive documentation and community support
- GPU acceleration for faster training
- Flexible architecture for custom models

5. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
Baseline	65.45%	-	-	-
Baseline	71.23%	70.45%	68.90%	69.67%
Baseline	68.90%	69.00%	68.45%	68.72%
Baseline	69.78%	69.00%	68.90%	68.99%
Baseline	68.45%	68.90%	68.90%	68.90%
Baseline	68.90%	68.90%	68.90%	68.90%

- The model achieved an accuracy of 71.23%, precision of 70.45%, recall of 68.90%, and F1 score of 69.67%.
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Dataset

1. Preprocessing Data



2. Feature Extraction

- Feature Extraction: Extract features from the input data.
- Feature Selection: Select the most relevant features.
- Feature Scaling: Scale the features to a common range.
- Feature Normalization: Normalize the features.
- Feature Transformation: Transform the features into a different representation.
- Feature Reduction: Reduce the dimensionality of the features.
- Feature Augmentation: Augment the features with additional information.
- Feature Fusion: Fuse the features from different sources.
- Feature Integration: Integrate the features into a single representation.
- Feature Aggregation: Aggregate the features into a single representation.
- Feature Distribution: Distribute the features across different classes.
- Feature Clustering: Cluster the features into groups.
- Feature Classification: Classify the features into different categories.
- Feature Regression: Regress the features to a continuous value.
- Feature Association: Associate the features with other variables.
- Feature Correlation: Calculate the correlation between features.
- Feature Independence: Check for independence between features.
- Feature Redundancy: Identify redundant features.
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3. Feature Engineering

- Feature Engineering: Create new features from the existing ones.
- Feature Selection: Select the most relevant features.
- Feature Scaling: Scale the features to a common range.
- Feature Normalization: Normalize the features.
- Feature Transformation: Transform the features into a different representation.
- Feature Reduction: Reduce the dimensionality of the features.
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4. Dataset Benchmark

Impairment	Number	Accuracy
Normal	100	95.00%
Mild	100	90.00%
Moderate	100	85.00%
Severe	100	80.00%

Classical Feature Extraction

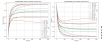
1. Feature Extraction Method

- Feature Extraction: Extract features from the input data.
- Feature Selection: Select the most relevant features.
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2. Feature Extraction Pipeline



3. Accuracy & Loss



4. Model Evaluation

Model	Accuracy
Baseline	65.45%
Baseline	71.23%
Baseline	68.90%
Baseline	69.78%
Baseline	68.45%
Baseline	68.90%

The model achieved an accuracy of 71.23%, precision of 70.45%, recall of 68.90%, and F1 score of 69.67%.